

Intron retention is a widespread mechanism of tumor-suppressor inactivation

Hyunchul Jung^{1–3}, Donghoon Lee^{1,4}, Jongkeun Lee^{1,5}, Donghyun Park^{2,6}, Yeon Jeong Kim^{2,6}, Woong-Yang Park^{2,7}, Dongwan Hong^{1,5}, Peter J Park^{8,9} & Eunjung Lee⁸

A substantial fraction of disease-causing mutations are pathogenic through aberrant splicing. Although genome profiling studies have identified somatic single-nucleotide variants (SNVs) in cancer, the extent to which these variants trigger abnormal splicing has not been systematically examined. Here we analyzed RNA sequencing and exome data from 1,812 patients with cancer and identified ~900 somatic exonic SNVs that disrupt splicing. At least 163 SNVs, including 31 synonymous ones, were shown to cause intron retention or exon skipping in an allele-specific manner, with ~70% of the SNVs occurring on the last base of exons. Notably, SNVs causing intron retention were enriched in tumor suppressors, and 97% of these SNVs generated a premature termination codon, leading to loss of function through nonsense-mediated decay or truncated protein. We also characterized the genomic features predictive of such splicing defects. Overall, this work demonstrates that intron retention is a common mechanism of tumor-suppressor inactivation.

Tumor-suppressor genes (TSGs) negatively regulate tumor progression by inhibiting cell division and differentiation, controlling DNA damage repair and immune responses, inducing apoptosis and suppressing metastasis¹. Loss or reduction in TSG function is a hallmark of cancer² and is frequently observed in various cancers. TSG inactivation occurs in several forms, including (i) protein-truncating variations such as nonsense, frameshift indel and splice-site mutations; (ii) structural variations³ such as genomic deletions and complex rearrangements; (iii) epigenetic silencing⁴ such as promoter hypermethylation and histone modification; and (iv) overexpression

of noncoding RNAs⁵ such as microRNAs, competing endogenous RNAs (ceRNAs) and pseudogenes, leading to post-transcriptional downregulation of TSG expression.

Somatic SNVs in classical splice sites (exon-intron junctions) and splicing regulatory elements, known as splicing enhancers and silencers, have been causally implicated in the dysregulation of several cancer-related genes^{6–9}. These mutations cause different forms of splicing aberration, such as exon skipping or intron retention. A recent report has described examples in which synonymous mutations result in alterations to exon usage and found that such mutations are enriched in oncogenes¹⁰. Although these examples suggest the importance of SNVs that disrupt splicing in cancer, how often such mutations arise and what type of splicing aberration they cause have not been analyzed. Given that SNVs that disrupt splicing are thought to be a primary pathogenic mechanism in many hereditary human diseases (accounting for an estimated 30–60% of disease-causing mutations^{11–14}), a more systematic analysis across a multitude of cancer samples is likely to be illuminating.

We carried out a comprehensive analysis of somatic SNVs resulting in aberrant splicing, using a large number of primary cancer specimens across six cancer types profiled in The Cancer Genome Atlas (TCGA) Project (**Supplementary Table 1**). For each of these samples, both whole-exome and mRNA sequencing (RNA-seq) data were available, allowing us to examine the mRNAs produced from the mutant alleles for splicing abnormality. Although previous studies have investigated mutational effect on splicing using both of these data types, they only analyzed a small set of such combined data (<20 cancer samples) and their analyses were limited to intronic splice-site mutations^{15,16}. Here we examined the potential impact of somatic SNVs identified from whole-exome data on splicing through analysis of matched RNA-seq data and examined the role of aberrant splicing in TSG inactivation.

RESULTS

Identification of somatic SNVs altering splicing

To identify somatic SNVs that disrupt splicing, we first compiled a total of 328,580 somatic SNVs on autosomes from 1,812 samples that underwent whole-exome sequencing (Online Methods). We excluded SNVs from genes that had additional SNVs or indels, to unambiguously trace the causal SNV for each splicing event. This filtering resulted in 230,311 SNVs, of which 31.6% were exonic mutations within 30 bp of exon-intron junctions and 1.5% were at intronic 5' and 3' splice sites (**Supplementary Table 1**).

¹Research Institute, National Cancer Center, Gyeonggi-do, South Korea.

²Samsung Genome Institute, Samsung Medical Center, Seoul, South Korea.

³Bioinformatics and Systems Biology Graduate Program, University of California, San Diego, La Jolla, California, USA. ⁴Program in Computational Biology and Bioinformatics, Yale University, New Haven, Connecticut, USA. ⁵Cancer Immunology Branch, Division of Cancer Biology, National Cancer Center, Gyeonggi-do, South Korea. ⁶Samsung Biomedical Research Institute, Samsung Advanced Institute of Technology, Samsung Electronics Company, Ltd., Seoul, South Korea. ⁷Department of Molecular Cell Biology, Sungkyunkwan University School of Medicine, Suwon, South Korea. ⁸Department of Medicine, Division of Genetics, Brigham and Women's Hospital, Boston, Massachusetts, USA. ⁹Department of Biomedical Informatics, Harvard Medical School, Boston, Massachusetts, USA. Correspondence should be addressed to E.L. (elee20@partners.org), P.J.P. (peter_park@hms.harvard.edu) or D.H. (dwhong@ncc.re.kr).

Received 30 November 2014; accepted 8 September 2015; published online 5 October 2015; doi:10.1038/ng.3414

In allele-specific splicing analysis, we detected SNVs causing intron retention (including ones resulting in intronic cryptic site activation) by identifying cases in which a mutant allele showed abnormal splicing whereas the wild-type (reference) allele supported normal splicing (Fig. 1a–d). For exonic SNVs close to exon-intron junctions, we were able to identify RNA-seq reads that reflected altered splicing and manifested the mutant allele at the same time. We then compared the proportion of such reads to the proportion of reads with the wild-type allele that showed abnormal splicing, using Fisher's exact test (Online Methods). For SNVs causing exon skipping, no reads with the mutant allele mapping to the skipped exon can be observed. We therefore looked for reads that spanned the junction of the two exons flanking the skipped exon with the SNV and filtered out SNVs with mutant allele expression to ensure the allele-specific occurrence of exon skipping (Online Methods; see **Supplementary Fig. 1** for a flowchart). This allele-specific splicing analysis identified 88 and 75 somatic exonic

SNVs that caused intron retention and exon skipping, respectively (see **Supplementary Table 2** for the list of SNVs).

Although allele-specific association of variation with aberrant splicing provides strong evidence for causality, it can miss some true events. For example, this approach fails to detect exon skipping that occurs at a moderate level, where some transcripts with the mutant SNV allele are still normally spliced. Thus, we also employed ratio-based splicing analysis where we calculated the ratio of abnormally spliced RNA-seq reads to normally spliced reads, as carried out previously^{17,18}, and evaluated whether the ratio was significantly higher than expected (Fig. 1e, Online Methods and **Supplementary Note**). The background distribution of ratios was estimated on the basis of ~2,500 RNA-seq data sets from normal tissues as well as cancer samples without any somatic mutations in the corresponding gene. To ensure that SNVs acted on splicing in *cis*, we excluded 62 cancer samples (407 SNVs) with mutations in *trans*-acting splicing factors, including *SF3B1* and *U2AF1* (Online Methods). Ratio-based splicing analysis resulted in the identification

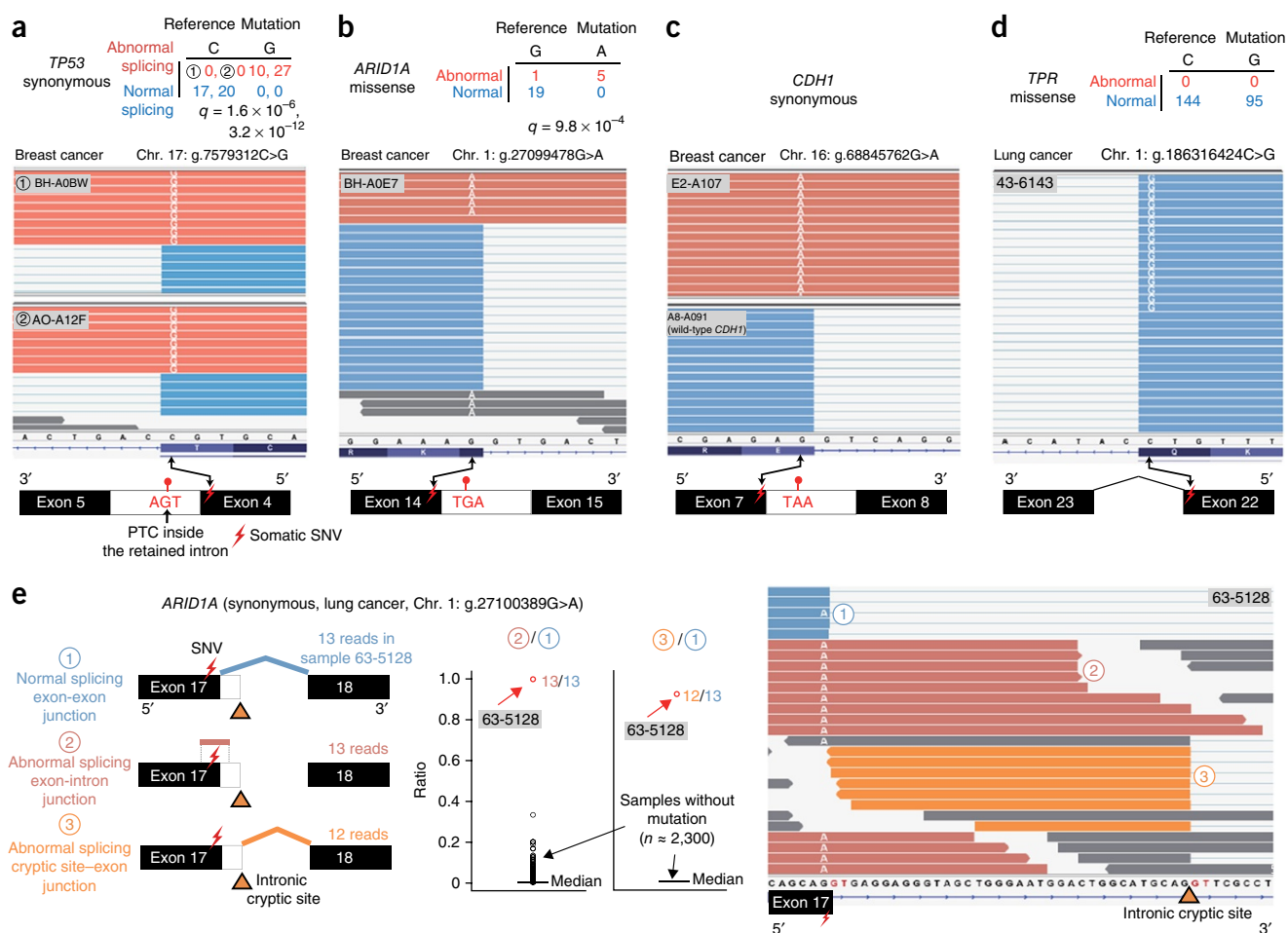
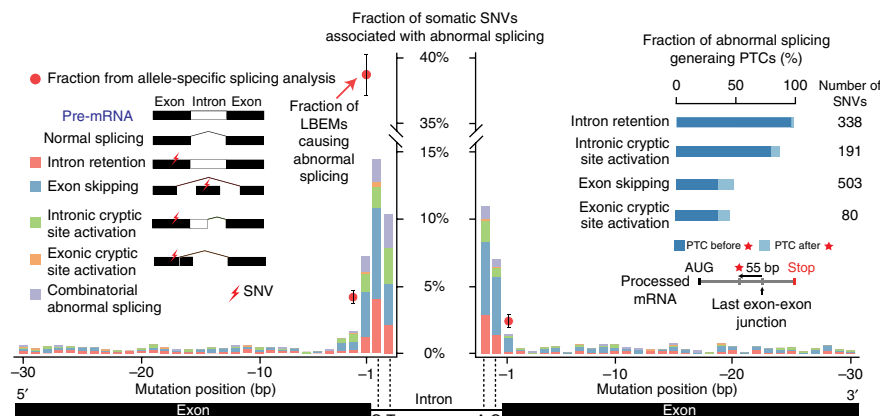


Figure 1 Identification of SNVs disrupting splicing. (a–c) Allele-specific splicing analysis of LBEMs causing intron retention. Reads showing intron retention or normal splicing are shown in red (mutant base in white) and blue, respectively. Reads that did not cover the mutant base or did not cover >10 bp of both sides of the exon-intron junction were not counted (gray). Not all reads are shown; sample IDs appear in gray boxes. Association between an SNV and splicing abnormality was tested using Fisher's exact test. (a) RNA-seq reads near a synonymous LBEM (lightning bolt) in *TP53* are shown for two breast cancer samples. The numbers for the two samples are separated by a comma. (b) Intron retention in *ARID1A* caused by a missense LBEM. (c) A synonymous LBEM causing intron retention in *CDH1*. A tumor sample with the LBEM (top) showed no transcripts with the reference allele, whereas another tumor sample without the LBEM (bottom) showed normal splicing. (d) An example of an LBEM that did not alter splicing. (e) Ratio-based splicing analysis of a synonymous LBEM causing intronic cryptic site activation in *ARID1A*. The number of RNA-seq reads supporting each of the normal (①) and abnormal (② and ③) splicing configurations was calculated from a cancer sample (63-5128). The two ratios of abnormally to normally spliced reads (②/① and ③/①) are shown in box plots (red dots). Background ratios from ~2,300 normal and cancer samples lacking mutations in *ARID1A* are shown as black dots. The intrinsic cryptic site (orange triangle) has a GT dinucleotide sequence identical to that of the 5' splice site.

Figure 2 Positional association of somatic SNVs with abnormal splicing.

Red dots mark the fraction of SNVs disrupting splicing at each position according to allele-specific splicing analysis (error bars, s.e.m.). The denominator of the fraction is the number of SNVs at a given position whose splicing effect could be confirmed by allele-specific splicing analysis. Dots are not presented for positions with a fraction <2%. A stacked bar shows the fraction of somatic SNVs associated with abnormal splicing identified by ratio-based splicing analysis (the fractions use the number of all SNVs at each position as the denominator).

The types of abnormal splicing we detected were intron retention, exon skipping, intronic or exonic cryptic site activation, and complex cases that showed multiple aberrations. The fraction of abnormal splicing events generating PTCs is shown for each abnormal splicing type (right inset).



of 1,200 somatic SNVs (848 exonic SNVs) associated with the five types of abnormal splicing (see **Fig. 2** and **Supplementary Fig. 2** for examples of complex abnormal splicing; **Supplementary Table 3**). Comparison of the two sets is shown in **Supplementary Figure 3**.

Frequently altered splicing by last-base exonic mutations

To evaluate the positional impact of SNVs on splicing, we assessed the fraction of abnormally spliced transcripts as a function of the distance of SNVs from the nearest exon-intron junction. Using the SNVs from allele-specific splicing analysis, we found that last-base exonic mutations (LBEMs) had the highest rate (38%; 114 of 299 LBEMs) of abnormal splicing among all exonic SNVs (**Fig. 2**). The remaining 185 LBEMs did not result in abnormal splicing (**Fig. 1d**), suggesting that their regulation of splicing involves other factors. The SNVs at penultimate and first bases of exons showed rates of approximately 4% and 3% for abnormal splicing, respectively (**Fig. 2**), marginally higher than the background rate (<2%).

Using the 1,200 SNVs from ratio-based splicing analysis, we consistently found that LBEMs resulted in the highest rate of abnormal splicing (~7%) among exonic loci, comparable to the rate for the second base of the 3' intronic splice site (~7%; **Fig. 2**). The last base of exons also showed a higher overall mutation frequency than the penultimate or first base (2.0% versus 1.5% and 1.2%; **Supplementary Table 4**). Because the last base of exons belongs to the classical 9-nt sequence motif of the 5' splice site recognized by the U1 small nuclear RNA (snRNA)¹⁹, LBEMs are likely to cause altered splicing by interrupting the binding of this snRNA to the 5' splice site.

Among the five types of abnormal splicing identified from ratio-based splicing analysis, exon skipping ($n = 503$) and intron retention ($n = 338$) were the most frequently observed. SNVs causing exon skipping were enriched in exonic splicing enhancers ($P = 8.3 \times 10^{-4}$) but not in exonic splicing silencers, whereas other SNVs such as those causing intron retention were not enriched in either category (**Supplementary Note**). This suggests that somatic SNVs disrupting exonic splicing enhancers frequently lead to exon skipping in cancer.

Mutational impact on gene expression

Abnormal splicing can trigger nonsense-mediated decay (NMD)—a surveillance process that eliminates mutant transcripts with a premature termination codon (PTC)²⁰. To understand the relationship between abnormal splicing and NMD, we examined whether each splicing defect generated a PTC and, if so, where the PTC was located. We found that 97% and 86% of SNVs causing intron retention and

intronic cryptic site activation, respectively, generated PTCs (**Fig. 2**). In contrast, SNVs causing exon skipping and exonic cryptic site activation showed much lower rates of PTC generation (48% and 45%, respectively). It is known that PTCs between ~55 bp upstream of the last exon-exon junction and the end of the last exon do not trigger NMD²⁰. Thus, we further examined the fraction of NMD-susceptible PTCs. We found that events involving introns showed a larger proportion of NMD-susceptible PTCs than those involving exons, suggesting prevalent NMD in intron-associated splicing defects (**Fig. 2** and **Supplementary Table 3**).

To understand the impact of somatic SNVs on gene expression through NMD, we computed the fraction of somatic SNVs at each position that resulted in expression of only the reference allele, using all 72,810 exonic SNVs within 30 bp of exon-intron junctions. As expected, nonsense SNVs, known to be subject to NMD, showed a higher fraction (~55% on average) of variants resulting in expression of only the reference allele than missense or synonymous SNVs (~32%). LBEMs (missense or synonymous) showed a remarkably high fraction (53%), comparable to that for nonsense SNVs (**Fig. 3a**, **Supplementary Fig. 4** and **Supplementary Table 5**). Given the high rates of intron retention and exon skipping for LBEMs (**Fig. 2**), it is likely that PTCs in frame-shifted transcripts due to such aberrations triggered NMD and led to the absence of expression of mutant alleles.

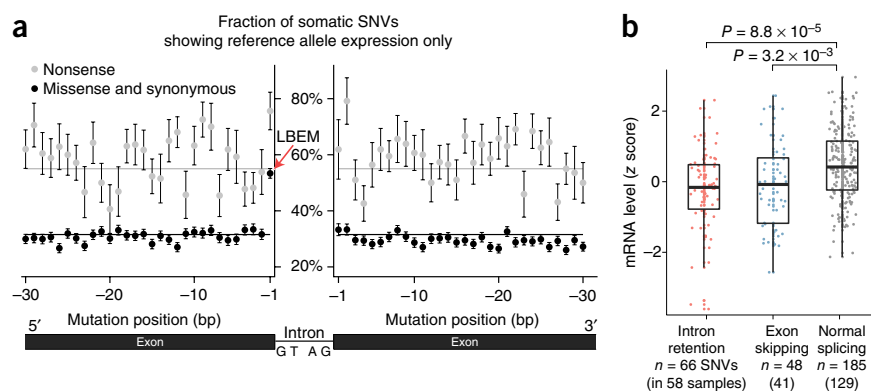
At the whole-transcript level, genes with LBEMs that caused intron retention or exon skipping according to allele-specific splicing analysis showed significantly lower expression than genes with LBEMs demonstrating normal splicing ($P = 8.8 \times 10^{-5}$ and 3.2×10^{-3} , respectively; **Fig. 3b**). The main mechanism underlying decreased expression for the genes with PTC-generating LBEMs is likely to be NMD, although some transcripts subject to NMD can escape degradation through a currently unknown mechanism²¹ and the efficacy of triggered NMD can be variable between samples^{22,23}.

Enrichment of intron retention-causing SNVs in TSGs

To measure the relevance of SNVs disrupting splicing in tumorigenesis, we tested whether they were enriched in oncogenes or TSGs (see **Fig. 4a** and **Supplementary Fig. 5** for SNVs from allele-specific and ratio-based splicing analysis, respectively; Online Methods). We found that SNVs causing intron retention were enriched in all four TSG sets but not in the oncogene sets we curated. However, SNVs causing exon skipping—even those at intronic splice sites—were not enriched in TSGs, except for a slight enrichment of intronic splice-site SNVs in one of the four TSG sets (**Fig. 4a**).

Figure 3 Disruption of gene expression by LBEMs. (a) Fraction of SNVs showing expression of only the reference allele. Gray and black dots represent the fractions of nonsense SNVs and other types of SNVs (synonymous and missense), respectively, that showed reference allele-only expression at each mutational position. Error bars, s.e.m. Solid horizontal lines show the average fraction across all exonic positions. (b) Change in mRNA expression from abnormal splicing. The z score corresponding to expression for each gene (normalized to have mean = 0 and s.d. = 1 across cancer samples for each tumor type) was plotted for each splicing group.

A Mann-Whitney *U* test was used to measure the statistical significance of the differences between groups. The horizontal line within each box shows the median, and the box spans the first quartile to the third quartile (interquartile range, IQR). Whiskers show $1.5 \times$ IQR.



Of the 66 LBEMs causing intron retention, 16 (24%), including seven in *TP53*, occurred in TSGs (Table 1). Among the patients with truncating mutations (nonsense or intronic splice-site SNVs, frameshift indels or LBEMs causing intron retention) in TSGs, 1–25% had an LBEM causing intron retention (Table 1 and Supplementary Table 6). When considering truncating SNVs alone, cases with this type of mutation comprised 2–33%. This suggests that LBEMs causing intron retention represent an important, previously underappreciated mechanism of TSG inactivation.

Through additional screening of cancer gene mutations (missense and synonymous SNVs), including the data from the Catalogue of Somatic Mutations in Cancer (COSMIC)²⁴ (Online Methods), we found 107 LBEMs that occurred in two or more patients, with 23 in known TSGs (Supplementary Table 7; note that RNA-seq data were

not examined). These 107 LBEMs also showed significant enrichment in the TSG sets ($P \leq 8.0 \times 10^{-5}$; Fig. 4a) but not in the oncogene sets.

Using minigene splicing assays²⁵, we experimentally validated that LBEMs in five TSGs—*ARID1A*, *PTEN*, *CDH1*, *MLL2* (*KMT2D*) and *PTCH1*—caused intron retention (Fig. 4b and Supplementary Fig. 6; Online Methods). For example, the transcript harboring the *ARID1A* LBEM (Fig. 1e) showed both full and partial (by intronic cryptic site activation) retention of the intron, whereas the wild-type *ARID1A* transcript showed predominantly normal splicing (only 5.8% of transcripts contained the target intron, likely owing to the presence of pre-mRNAs; Fig. 4b). Sanger sequencing confirmed the exact sequences of the expected junctions. This validation demonstrates that our computational analysis using paired DNA sequencing and RNA-seq data is effective in finding SNVs that alter splicing.

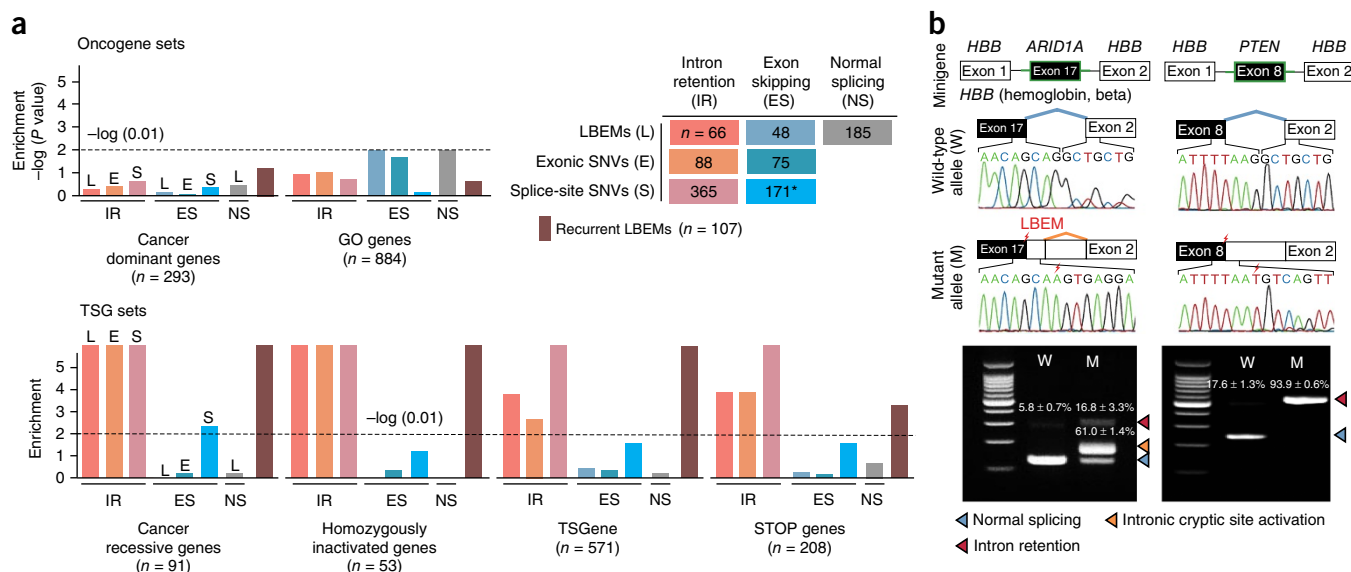


Figure 4 SNVs disrupting splicing in TSGs. (a) Enrichment in TSGs. Somatic SNVs (synonymous and missense) identified by allele-specific splicing analysis were tested for their enrichment in two oncogene sets (top) and four TSG sets (bottom). Because SNVs at intronic splice sites that cause exon skipping (marked by an asterisk) cannot be detected by allele-specific splicing analysis (Online Methods), variants from ratio-based splicing analysis were used (enrichment based on other SNVs from ratio-based splicing analysis is shown in Supplementary Fig. 5). *P* values were assessed by random permutation test. Bars exceeding the y-axis value of 5 represent <1 observation in 100,000 random trials. (b) Experimentally confirmed LBEMs in TSGs causing intron retention. Validation experiments for the LBEM in exon 17 of *ARID1A* (RNA-seq reads shown in Fig. 1e) and the LBEM in exon 8 of *PTEN* are shown. The whole exon with an LBEM (green box) and flanking intronic sequences (~90–130 bp; green lines) of wild-type and mutant alleles were cloned and used to replace an alternative cassette in a reporter plasmid. On the corresponding gels, red and orange triangles indicate transcripts with full and partial intron retention, respectively. The percentage of abnormally spliced transcripts is shown above each band (mean $\pm$ s.d. percentage from triplicate analysis).

Table 1 Tumor-suppressor genes with LBEMs causing intron retention

Tumor type	Confirmed intron retention		Predicted intron retention		Samples with mutations (% truncating mutations, % LBEMs)
	Silent	Missense	Silent	Missense	
Breast invasive carcinoma (<i>n</i> = 503)	<i>MLL3</i> ^a <i>TP53</i> (2)	<i>ARID1A</i>	<i>CDH1</i>		<i>ARID1A</i> (2%, 12%) <i>CDH1</i> (6%, 3%) <i>MLL3</i> (4%, 5%) <i>TP53</i> (14%, 3%)
Colorectal carcinoma (<i>n</i> = 217)	<i>TP53</i>			<i>TP53</i> <i>VPS13A</i>	<i>TP53</i> (18%, 5%)
Kidney renal clear cell carcinoma (<i>n</i> = 400)	<i>SETD2</i>	<i>CDKN2A</i> <i>SPSB3</i> (2) ^b <i>VHL</i> (2)		<i>FAT1</i>	<i>VHL</i> (30%, 2%) <i>SETD2</i> (8%, 3%)
Lung squamous cell carcinoma (<i>n</i> = 178)	<i>ARID1A</i> <i>TP53</i> (3)	<i>CDKN2A</i> (exon 2) <i>CIC</i> <i>MLL2</i> , <i>MLL2</i> ^a <i>PSMC3</i> <i>TP53</i> (2) <i>WRN</i>	<i>TP53</i> (2)	<i>CDKN2A</i> (exon 1) <i>LRP1B</i> <i>PTCH1</i>	<i>ARID1A</i> (4%, 14%) <i>CDKN2A</i> (8%, 14%) <i>LRP1B</i> (5%, 12%) <i>MLL2</i> (12%, 10%) <i>TP53</i> (28%, 14%)
Ovarian serous cystadenocarcinoma (<i>n</i> = 273)	<i>TP53</i>				<i>TP53</i> (34%, 1%)
Uterine corpus endometrial carcinoma (<i>n</i> = 241)		<i>BAP1</i> <i>PTCH1</i> <i>KDM5C</i>		<i>PTEN</i>	<i>PTCH1</i> (2%, 25%) <i>PTEN</i> (47%, 1%)

TSGs with LBEMs causing intron retention are shown for each cancer type. The number of patients with LBEMs for each TSG is shown in parentheses when there was more than one patient. LBEMs were either confirmed by allele-specific RNA-seq read pattern analysis, including further manual inspection (**Supplementary Table 6**), or predicted by our computational model. The prediction model was applied to LBEMs that did not meet the expression requirements for allele-specific splicing analysis (cases with expression of only the reference allele, only the alternate allele or both alleles at levels that were too low). We manually inspected RNA-seq read patterns for LBEMs that were excluded from splicing analysis (for example, those occurring with multiple variants in the same gene and those on sex chromosomes). The percentage of all cancer samples with truncating mutations (nonsense or splice-site SNVs, frameshift indels or SNVs causing intron retention) and the percentage of the samples with truncating mutations having SNVs causing intron retention are shown in parentheses. LBEMs with <1% in either category are not shown.

^aSNV at the penultimate base of an exon. ^b*SPSB3* is shown, although it is not a known tumor suppressor, because of the occurrence of the same LBEM in this gene in two patients.

TP53 was the most frequently disrupted gene among the genes with LBEMs causing intron retention (with nine synonymous and three missense variants; **Table 1**). Interestingly, all these LBEMs mapped to only three different exons (exons 4, 6 and 9; **Supplementary Fig. 7**), suggesting that the last bases of these three exons are functional hotspots for *TP53* inactivation through intron retention. These three loci in *TP53* were recently reported to have recurrent synonymous SNVs¹⁰. In lung squamous cell carcinoma (LUSC), for example, four of 178 patients carried no mutation in *TP53* other than one of the three synonymous LBEMs; taking these mutations into account increases the percentage of patients with defective *TP53* by ~3% (from 79% to 82%). Among the LBEMs occurring in genes previously not recognized as TSGs, a missense LBEM in exon 6 of *SPSB3* caused intron retention in two patients with kidney cancer (**Table 1** and **Supplementary Fig. 8**). The fact that *SPSB3* interacts with a proto-oncogene, the *MET* receptor tyrosine kinase²⁶, and that high *SPSB3* expression correlates with favorable prognosis in non-small cell lung cancer²⁷ suggests a possible disruption of the TSG function of *SPSB3* due to intron retention caused by the LBEM.

TSG inactivation by intron retention

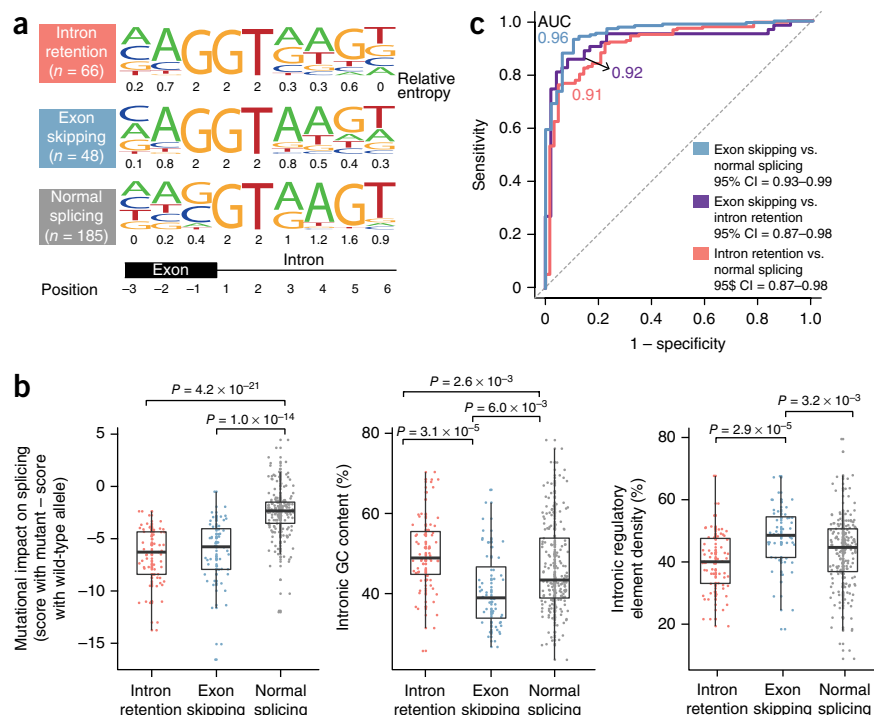
Expression analysis of TSGs with intron retention recapitulated the mechanisms leading to loss of function: (i) an insufficient level of TSG protein due to NMD after PTC generation²⁸ or (ii) truncated TSG protein generated from a transcript with a PTC that escaped NMD (for an unknown reason)²⁰ (**Supplementary Fig. 9**). For example, an LBEM in *TP53* in sample BH-A0BW (**Fig. 1a**, top) resulted in a decrease in gene expression comparable to that observed with typical truncating mutations (**Supplementary Fig. 10a**). The same LBEM in *TP53* in sample AO-A12F (**Fig. 1a**, bottom) also caused intron retention, but this sample showed intact mRNA levels, indicating variable NMD efficacy among individuals^{22,23} (**Supplementary Fig. 10a**). In this case, the LBEM may inactivate *TP53* through protein truncation.

Allelic expression of TSGs provided further insights on TSG inactivation. For example, samples with haploinsufficiency for TSGs such as *TP53* (ref. 29), *ARID1A*³⁰ (**Fig. 1a,b**) and *VHL*³¹ (**Table 1**) with intron retention showed intact expression of wild-type alleles, demonstrating partial inactivation. In contrast, other TSGs such as *CDH1* (**Fig. 1c** and **Supplementary Fig. 6**) and *PTEN* (**Fig. 4b** and **Table 1**) showed complete inactivation, as demonstrated by the complete absence of wild-type allele expression, likely due to other mechanisms such as DNA copy loss or DNA methylation. Indeed, loss of the *CDH1* allele without intron retention was confirmed by array-based copy number profiling by TCGA. Our filtering for one SNV per gene might preclude the analysis of TSGs like *CDH1* that follow the 'two-hit hypothesis' of biallelic inactivation. Thus, for known TSGs, we relaxed this requirement and found two additional LBEMs causing intron retention in TSGs: a missense SNV in *BAP1* and a synonymous SNV in *TP53* (**Table 1**). Mutations in the same TSG led to unique inactivation patterns. For example, *CDKN2A* with an LBEM in exon 2 causing intron retention showed partial inactivation, whereas the same TSG with an LBEM in exon 1 causing intron retention showed complete inactivation (**Table 1**). Additional experimental and functional characterization will be needed to better understand the context-dependent mode of inactivation for individual events.

Characterization and prediction of LBEMs altering splicing

We earlier showed that LBEMs cause different forms of splicing aberration but also identified an LBEM that did not disrupt splicing (**Fig. 1d**). To understand the factors determining the potential of LBEMs to disrupt splicing, we compared the genomic features of these mutation sites across different abnormal splicing groups. First, we looked for informative motifs in the local sequence of LBEMs from allele-specific splicing analysis (**Fig. 5a**). In the abnormal splicing groups, the two exonic bases from the exon-intron junction were more conserved than the intronic bases, as demonstrated by their higher relative entropy scores³², except for the universally conserved

Figure 5 Characterization of discriminative features for splicing aberration and construction of prediction models. **(a)** Distinct LBEM sequence motifs. LBEMs in different splicing groups show distinct sequence motifs near exon-intron junctions. The motifs were derived from the nine wild-type bases (three exonic and six intronic) of the 5' donor splice sites. The relative entropy score is shown for each position. **(b)** Examples of features predictive for splicing aberration. The change in splicing strength exerted by LBEMs, local GC content and the density of intronic splicing regulatory elements was compared among the three splicing groups. A Mann-Whitney *U* test was used to measure the statistical significance of the differences. **(c)** Accuracy of our prediction models. A ROC curve along with the AUC value is depicted for each of the three prediction models.



canonical GT dinucleotides. In the group with normal splicing, the trend was the opposite. LBEMs that disrupt splicing or do not affect splicing mirror two distinctive groups of 5' splice-site motifs previously reported: 'left-handed' and 'right-handed' models with higher relative entropy in the exon and in the intron, respectively. These motifs are known to be regulated by different splicing factors such as U5 (left handed) and U6 (right handed) snRNAs, respectively^{33,34}. LBEMs disrupting splicing decreased the strength of splicing more than LBEMs that did not disrupt splicing ($P \leq 1.0 \times 10^{-14}$; see Fig. 5b and Supplementary Fig. 11 for MAXENT³³ and H-bond scores, respectively).

The genomic features that distinguish exon skipping and intron retention correspond to the characteristics of two major splicing models: exon definition, where exons are used as initial recognition sites by splicing factors, and intron definition, where introns are used as recognition sites by splicing factors, respectively^{6,35–37}. Major discriminative features include a higher GC content and a lower density of intronic splicing regulatory elements in intron retention in comparison to exon skipping (Fig. 5b and Supplementary Figs. 12–14; additional descriptions of the features are presented in the Supplementary Note). Thus, LBEMs under the left-handed motif context are likely to result in exon skipping if the LBEM site was originally governed by exon definition and intron retention if the site was originally governed by intron definition.

We built classification models to predict the type of abnormal splicing given an LBEM using Lasso logistic regression³⁸. In addition to the genomic features characterized above, we included transcriptomic features for classification (Fig. 5c, Supplementary Fig. 15 and Supplementary Table 8). Splicing strength scores and exon expression levels were the most discriminative features (see Supplementary Table 9 for features ranked by discriminative power). Our leave-one-out cross-validation (LOOCV) showed high accuracy of the models, with an area under the receiver operating characteristic (ROC) curve (AUC) of 0.92–0.96 (Fig. 5c).

By applying the prediction models to 474 LBEMs that were not qualified for allele-specific splicing analysis (having insufficient expression of either allele), we identified 125 and 65 additional LBEMs that are likely to cause intron retention and exon skipping, respectively (Table 1 and Supplementary Table 10). The LBEMs predicted to disrupt splicing included ones without detectable levels

of mutant transcripts (perhaps because of the complete removal of these transcripts by NMD). For example, an LBEM in *PTCH1*, a known haploinsufficient TSG that negatively regulates Hedgehog signaling³⁹, in a patient with LUSC was predicted to cause intron retention, although mutant transcripts were completely absent in the RNA-seq data (Table 1). The predicted set of LBEMs showed sequence motif and gene set enrichment patterns consistent with previous analyses using the LBEMs from allele-specific splicing analysis (Supplementary Fig. 16). Without using transcriptomic features (as RNA-seq data are often unavailable), the prediction of intron retention versus normal splicing showed identical performance. In this scenario, prediction performance decreased by ~8% when predicting exon skipping versus normal splicing and by ~14% when predicting exon skipping versus intron retention.

DISCUSSION

Our analysis of paired DNA sequencing and RNA-seq data across a large number of cases has showed the critical role of somatic SNVs, especially LBEMs, in disrupting splicing in primary tumors. Following on a previous analysis of ESTs suggesting that altered splicing might disrupt TSG function⁴⁰, our comprehensive analysis found a large set of splicing aberrations, traced the source of each aberration to a specific somatic SNV, characterized splicing defects in detail and showed enrichment of intron retention in TSGs across cancer types.

The greater frequency of intron retention for TSG inactivation in comparison to exon skipping can be explained by the fact that the majority of observed intron retention events (~97%) generated PTCs whereas only 50% of the exon skipping events did (Fig. 2), and a disrupted reading frame may be more prone to induce loss of function in TSGs than gain of function in oncogenes. Our explanation is supported by the fact that in-frame synonymous exonic SNVs disrupting exon usage were enriched in oncogenes but not in TSGs¹⁰, and in-frame exon skipping is known to activate several oncogenes such as *MET*⁴¹.

This study highlights the importance of synonymous as well as missense somatic SNVs in TSG inactivation through disruption

of splicing, with these variants showing a potency for inactivation similar to that for traditional nonsense SNVs. This result suggests that somatic SNVs beyond intronic splice sites should be carefully considered for their potential as disease-causing variants, regardless of whether an amino acid change occurs. Finally, even when matched RNA-seq data are not available or have limited signal owing to NMD, our prediction models can accurately annotate LBEMs for their potential to disrupt splicing. This analysis is likely to prove beneficial as part of the standard analysis pipeline for exome and genome analysis, both for cancer and other diseases.

URLs. Oncotator, <http://www.broadinstitute.org/oncotator/>; Picard, <http://broadinstitute.github.io/picard/>.

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

ACKNOWLEDGMENTS

We thank S.J. Elledge, H. Schaa, A. Han, N. Gehlenborg and N. Smedemark-Margulies for insightful discussion on tumor-suppressor inactivation, providing a script to determine H-bond scores, advice on minigene experimental validation, comments on visualization, and critical reading and editing of the manuscript, respectively. This study was supported by a grant from the National Cancer Center, Korea (NCC-1310190 and NCC-1410675 to D.H.). E.L. was supported by the Harvard Medical School Eleanor and Miles Shore fellowship and William Randolph Hearst Fund.

AUTHOR CONTRIBUTIONS

H.J., P.J.P. and E.L. designed the study and analyzed the data. H.J., D.L. and J.L. performed bioinformatics analyses. D.P., Y.K. and W.P. performed minigene experiments. H.J., D.L., P.J.P. and E.L. wrote the manuscript. D.H., P.J.P. and E.L. supervised the project.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

Reprints and permissions information is available online at <http://www.nature.com/reprints/index.html>.

- Sherr, C.J. Principles of tumor suppression. *Cell* **116**, 235–246 (2004).
- Hanahan, D. & Weinberg, R.A. Hallmarks of cancer: the next generation. *Cell* **144**, 646–674 (2011).
- Zack, T.I. *et al.* Pan-cancer patterns of somatic copy number alteration. *Nat. Genet.* **45**, 1134–1140 (2013).
- Shen, H. & Laird, P.W. Interplay between the cancer genome and epigenome. *Cell* **153**, 38–55 (2013).
- Esteller, M. Non-coding RNAs in human disease. *Nat. Rev. Genet.* **12**, 861–874 (2011).
- Wang, Z. & Burge, C.B. Splicing regulation: from a parts list of regulatory elements to an integrated splicing code. *RNA* **14**, 802–813 (2008).
- Baralle, D. & Baralle, M. Splicing in action: assessing disease causing sequence changes. *J. Med. Genet.* **42**, 737–748 (2005).
- Venables, J.P. Aberrant and alternative splicing in cancer. *Cancer Res.* **64**, 7647–7654 (2004).
- Kalnina, Z., Zayakin, P., Silina, K. & Line, A. Alterations of pre-mRNA splicing in cancer. *Genes Chromosom. Cancer* **42**, 342–357 (2005).
- Supek, F., Minana, B., Valcarcel, J., Gabaldon, T. & Lehner, B. Synonymous mutations frequently act as driver mutations in human cancers. *Cell* **156**, 1324–1335 (2014).
- López-Bigas, N., Audit, B., Ouzounis, C., Parra, G. & Guigo, R. Are splicing mutations the most frequent cause of hereditary disease? *FEBS Lett.* **579**, 1900–1903 (2005).
- Faustino, N.A. & Cooper, T.A. Pre-mRNA splicing and human disease. *Genes Dev.* **17**, 419–437 (2003).
- Cartegni, L., Chew, S.L. & Krainer, A.R. Listening to silence and understanding nonsense: exonic mutations that affect splicing. *Nat. Rev. Genet.* **3**, 285–298 (2002).
- Lim, K.H., Ferraris, L., Filloux, M.E., Raphael, B.J. & Fairbrother, W.G. Using positional distribution to identify splicing elements and predict pre-mRNA processing defects in human genes. *Proc. Natl. Acad. Sci. USA* **108**, 11093–11098 (2011).
- Liu, J. *et al.* Genome and transcriptome sequencing of lung cancers reveal diverse mutational and splicing events. *Genome Res.* **22**, 2315–2327 (2012).
- Spinelli, R. *et al.* Identification of novel point mutations in splicing sites integrating whole-exome and RNA-seq data in myeloproliferative diseases. *Mol. Genet. Genomic Med.* **1**, 246–259 (2013).
- Xiong, H.Y. *et al.* RNA splicing. The human splicing code reveals new insights into the genetic determinants of disease. *Science* **347**, 1254806 (2015).
- Danan-Gothold, M. *et al.* Identification of recurrent regulated alternative splicing events across human solid tumors. *Nucleic Acids Res.* **43**, 5130–5144 (2015).
- Roca, X., Krainer, A.R. & Eperon, I.C. Pick one, but be quick: 5' splice sites and the problems of too many choices. *Genes Dev.* **27**, 129–144 (2013).
- Maquat, L. *Nonsense-Mediated mRNA Decay* (Landes Bioscience, 2006).
- Lappalainen, T. *et al.* Transcriptome and genome sequencing uncovers functional variation in humans. *Nature* **501**, 506–511 (2013).
- Nguyen, L.S., Wilkinson, M.F. & Gecz, J. Nonsense-mediated mRNA decay: inter-individual variability and human disease. *Neurosci. Biobehav. Rev.* **46**(pt. 2), 175–186 (2014).
- Huang, L. & Wilkinson, M.F. Regulation of nonsense-mediated mRNA decay. *Wiley Interdiscip. Rev. RNA* **3**, 807–828 (2012).
- Forbes, S.A. *et al.* COSMIC: mining complete cancer genomes in the Catalogue of Somatic Mutations in Cancer. *Nucleic Acids Res.* **39**, D945–D950 (2011).
- Cooper, T.A. Use of minigene systems to dissect alternative splicing elements. *Methods* **37**, 331–340 (2005).
- Wang, D., Li, Z., Messing, E.M. & Wu, G. The SPRY domain-containing SOCS box protein 1 (SSB-1) interacts with MET and enhances the hepatocyte growth factor-induced Erk-Elk-1-serum response element pathway. *J. Biol. Chem.* **280**, 16393–16401 (2005).
- Van Laar, R.K. Genomic signatures for predicting survival and adjuvant chemotherapy benefit in patients with non-small-cell lung cancer. *BMC Med. Genomics* **5**, 30 (2012).
- Tazi, J., Bakkour, N. & Stamm, S. Alternative splicing and disease. *Biochim. Biophys. Acta* **1792**, 14–26 (2009).
- Berger, A.H., Knudson, A.G. & Pandolfi, P.P. A continuum model for tumour suppression. *Nature* **476**, 163–169 (2011).
- Wu, J.N. & Roberts, C.W. *ARID1A* mutations in cancer: another epigenetic tumor suppressor? *Cancer Discov.* **3**, 35–43 (2013).
- Bader, H.L. & Hsu, T. Systemic *VHL* gene functions and the VHL disease. *FEBS Lett.* **586**, 1562–1569 (2012).
- Lawrence, C.E. *et al.* Detecting subtle sequence signals: a Gibbs sampling strategy for multiple alignment. *Science* **262**, 208–214 (1993).
- Yeo, G. & Burge, C.B. Maximum entropy modeling of short sequence motifs with applications to RNA splicing signals. *J. Comput. Biol.* **11**, 377–394 (2004).
- Burge, C.B., Tuschl, T. & Sharp, P.A. in *The RNA World: The Nature of Modern RNA Suggests a Prebiotic RNA World* 2nd edn. Vol. 37 (ed. Gesteland, R.F.) 525–560 (Cold Spring Harbor Laboratory Press, 1993).
- Keren, H., Lev-Maor, G. & Ast, G. Alternative splicing and evolution: diversification, exon definition and function. *Nat. Rev. Genet.* **11**, 345–355 (2010).
- Ast, G. How did alternative splicing evolve? *Nat. Rev. Genet.* **5**, 773–782 (2004).
- Amit, M. *et al.* Differential GC content between exons and introns establishes distinct strategies of splice-site recognition. *Cell Rep.* **1**, 543–556 (2012).
- Friedman, J., Hastie, T. & Tibshirani, R. Regularization paths for generalized linear models via coordinate descent. *J. Stat. Softw.* **33**(1), 1–22 (2010).
- Yang, L., Xie, G., Fan, Q. & Xie, J. Activation of the hedgehog-signaling pathway in human cancer and the clinical implications. *Oncogene* **29**, 469–481 (2010).
- Xu, Q. & Lee, C. Discovery of novel splice forms and functional analysis of cancer-specific alternative splicing in human expressed sequences. *Nucleic Acids Res.* **31**, 5635–5643 (2003).
- Cancer Genome Atlas Research Network. Comprehensive molecular profiling of lung adenocarcinoma. *Nature* **511**, 543–550 (2014).

ONLINE METHODS

Exome and transcriptome data sets and preprocessing. We generated a list of 1,812 cancer samples for six cancer types, profiled with both exome sequencing and RNA-seq, from six published papers by TCGA^{42–47} (**Supplementary Table 1**). We obtained somatic SNVs from the original literature and downloaded RNA-seq bam files from the Cancer Genomics Hub (CGHub; accessed June 2013). RNA-seq read lengths were in the range of 50–75 bp (**Supplementary Table 1**). The hg18 coordinates of SNVs from TCGA colorectal and ovarian cancer samples were converted to hg19 coordinates using the UCSC liftOver tool⁴⁸. All somatic SNVs were annotated with canonical RefSeq transcripts⁴⁹ using Oncotator. To assess the direct effect of an SNV on pre-mRNA splicing and gene expression, we filtered out genes with multiple somatic SNVs from each sample. In addition, genes with a single exon were excluded from further analysis, as they were irrelevant to the study of splicing. We focused only on target exons with canonical GT-AG splice-site pairs and their adjacent exons. The distance of an SNV position to 5' and 3' splice sites was calculated using BEDTools⁵⁰ and a custom Python script. To avoid issues of zygosity related to sex chromosomes when studying allele-specific expression and splicing effects, we focused only on SNVs on autosomes.

To achieve more accurate RNA-seq read mapping for allele-specific splicing analysis, we extracted the reads aligned to genes with somatic SNVs from RNA-seq bam files and realigned them to the hg19 human reference genome using GSNAP, an alignment tool known to be effective in the mapping of split junctions (default settings with `--novelsplicing = 1`) (ref. 51). For paired-end sequencing data, if at least one read from a pair mapped to a gene containing an SNV, both reads were extracted for realignment. Hard-clipped reads and reads marked as PCR duplicates or non-primary alignment were discarded. Furthermore, reads with mapping quality scores less than 20 or with indels were filtered out.

Allele-specific splicing analysis. Our allele-specific splicing analysis requires significant association between SNV alleles and splicing patterns. For exonic SNVs, this analysis can detect intron retention (including intronic cryptic site activation) and exon skipping. For intronic (5' and 3') splice-site SNVs, this analysis can only detect intron retention, as this is the only way that intronic loci appear in RNA-seq reads. Throughout the analysis, we used the following criteria to label categories of allelic expression: expression of both alleles (≥ 2 reads each for both the reference and alternative alleles), expression of only the reference allele (≥ 2 reference allele reads and ≤ 1 alternative allele read), expression of only the mutant allele (≥ 2 alternative allele reads and ≤ 1 reference allele read) and no expression (both alleles having ≤ 1 read).

Detection of exonic SNVs causing intron retention. For each of 72,810 somatic exonic SNVs within 30 bp of an exon-intron junction, we extracted reads mapping to the region around the SNV and having base quality ≥ 20 at the SNV locus and mapping quality ≥ 20 from the realigned RNA-seq bam file. For each of the extracted RNA-seq reads, we examined which SNV allele (reference or mutant) it contained. Also, we determined whether the read supported normal splicing or abnormal splicing by the following rules. Let E*, I, C and E represent the exon containing the somatic SNV, the adjacent intron, an intronic cryptic splicing site in the adjacent intron and the adjacent exon, respectively. Upstream and downstream sequences were investigated separately. The read was considered to support (i) normal splicing if it spanned the E*-E junction and covered ≥ 10 bp of both exons and (ii) abnormal splicing if it spanned the E*-I junction and covered ≥ 10 bp of both E* and I (intron retention) or if it spanned the C-E junction and covered ≥ 10 bp of E (intronic cryptic site activation; examples in **Supplementary Fig. 17**). Then, we counted the number of reads for each of the four categories depending on the SNV allele (reference versus mutant) and the splicing status (normal versus abnormal), and we tested the association between the SNV and splicing using Fisher's exact test (see examples in **Fig. 1a–d** and **Supplementary Fig. 18**). For accurate statistical testing, SNVs had to have ≥ 3 reads in ≥ 2 categories in the Fisher's exact test.

P values from Fisher's exact test were grouped by mutation position relative to the nearest splice site and corrected by the Benjamini-Hochberg multiple-testing correction method⁵² for each group to adjust for a different background distribution at each position. Finally, somatic SNVs with $q < 0.1$ where there

were more mutant allele reads showing abnormal splicing than reference allele reads showing normal splicing were classified as intron retention-causing SNVs, whereas those with $q \geq 0.1$ where there were fewer mutant allele reads showing abnormal splicing than reference allele reads showing normal splicing were classified as SNVs not affecting splicing. We also note that Fisher's testing relying on RNA-seq read counts is likely to underdetect splicing-disrupting SNVs with low expression as compared to those with high expression.

Detection of exonic SNVs causing exon skipping. If an exonic SNV induces complete exon skipping, we will not observe that SNV in any mature mRNAs—instead, we will only observe RNA-seq reads spanning the two flanking exons. In this case, we cannot use Fisher's exact test. Instead, we performed ratio-based splicing analysis to look for unusual read patterns supporting exon skipping near exonic SNVs. From the SNVs that passed ratio-based splicing analysis, we selected SNVs with reference allele-only expression to select for candidates with a strong effect.

Detection of intronic splice-site SNVs causing intron retention. If a somatic SNV at an intronic splice site causes intron retention (including intronic cryptic site activation), the mutant allele can be observed in mRNAs retaining the intron, but the wild-type allele cannot be observed because the intron is normally spliced out (see examples in **Supplementary Fig. 19**). Thus, we selected intronic splice-site SNVs with mutant allele-only expression as candidates (see **Supplementary Table 11** for a list of SNVs). Reads supporting intron retention (including intronic cryptic site activation) were considered to support intron retention-causing SNVs (using the same criteria as for exonic SNVs causing intron retention). We cannot detect SNVs causing exon skipping because neither the wild-type nor mutant allele at intronic splice sites appears in RNA-seq reads unless the intron is retained.

Ratio-based splicing analysis. Our ratio-based splicing analysis detects somatic SNVs associated with different types of abnormal splicing: intron retention, exon skipping, and intronic and exonic cryptic site activation. SNVs that showed more than one abnormal splicing type were classified as complex abnormal splicing. Briefly, after removing PCR duplicate reads and reads with mapping quality < 20 or indels in the original alignment data (PCR duplication was marked using Picard), we classified each SNV as a synonymous, missense, nonsense or intronic splice-site variant and examined the junction alignment pattern of RNA-seq reads in the surrounding exonic regions, including the mutant exon and its adjacent up- and downstream exons. We calculated the ratio of abnormally spliced reads to normally spliced reads to assess the proportion of abnormal splicing. The criteria for each type of abnormal splicing are described in the **Supplementary Note**. To measure the effect of a mutant SNV on splicing, we first used all alternatively spliced transcripts in RefSeq to define 'normal' splicing patterns. Second, because RefSeq annotation is incomplete, we derived an additional set of normal splicing patterns using a large number of normal tissues (714) as well as tumor samples (up to ~1,800) that did not have any mutation in a given gene and had ≥ 2 reads supporting normal splicing (see **Supplementary Table 12** for a list of samples). Specifically, as ratio-based splicing analysis is based on the ratio between the numbers of reads supporting abnormal and normal splicing, we computed a background distribution for these ratios based on the above control samples. Then, for a somatic SNV to be associated with abnormal splicing, we required its observed ratio to be within the top 1% and 5% of the background distributions from the TCGA normal and tumor samples, respectively (see examples in **Fig. 1e**). To focus on abnormal splicing caused by *cis*-acting SNVs, we excluded SNVs from tumor samples ($n = 62$; **Supplementary Table 13**) with mutations in five *trans*-acting splicing factors (*PCBP1*, *SF3B1*, *SRSF2*, *U2AF1* and *ZRSR2*) recurrently affected in cancer^{53,54}.

Gene expression analysis. SNVs were classified on the basis of their allelic expression pattern: expression of both alleles, expression of only reference allele, expression of only alternate allele and no expression (see the criteria above). Allelic expression was estimated on the basis of realignment data using GSNAP. To estimate gene expression levels, reads per kilobase per million reads (RPKM)-based gene expression data were downloaded from the TCGA data portal (accessed January 2014). To estimate relative expression (log ratio) of the target exon containing an SNV and its adjacent exons, RNA-seq read coverage per exon was calculated using BEDTools⁵⁰.

Cancer gene set enrichment analysis. *Gene sets.* To assess whether splicing-disrupting somatic SNVs were enriched in cancer genes, we performed gene set enrichment analysis using permutation testing. The two oncogene sets we used comprised (i) 293 cancer dominant genes defined by COSMIC⁵⁵ and (ii) 884 GO (growth enhancer and oncogenes) genes that promoted cell proliferation in a short hairpin RNA (shRNA)-based screen⁵⁶. The four TSG sets tested comprised (i) 91 cancer recessive genes from COSMIC⁵⁵, (ii) 53 homozygously inactivated genes⁵⁶, (iii) 571 TSGs curated from the literature⁵⁷ and (iv) 208 STOP (suppressors of tumorigenesis and proliferation) genes that inhibited cell proliferation in an shRNA-based screen⁵⁶. Because only 15,567 autosomal genes were relevant to our analysis (on the basis of criteria such as having multiple exons; **Supplementary Table 1**), we removed excluded genes from the oncogene and TSG sets.

Permutation testing. For the groups of SNVs classified as type *i* (LBEMs, exonic SNVs or splice-site SNVs) and type *j* (SNVs resulting in intron retention, exon skipping or normal splicing), we counted how often N_{ij} splicing-altering SNVs occurred in a cancer gene in the cancer gene set. To evaluate whether the observed number X_{ij} was significantly different from the expected number, we generated a null distribution by randomly selecting N_{ij} SNVs from 213,729 missense and synonymous somatic SNVs (from the original total set; **Supplementary Table 1**) 100,000 times. The *P* value was calculated as the fraction of trials in which the occurrence of N_{ij} randomly selected SNVs in a gene in the cancer gene set was greater than or equal to X_{ij} .

Cancer mutation data set. To identify recurrent LBEMs, we first downloaded COSMIC v68 data²⁴ and removed duplicate entries with the same sample ID. After discarding SNVs from TCGA samples in the COSMIC data, we merged the data with the previously published TCGA somatic SNV data⁵⁸. We then selected synonymous or missense LBEMs that occurred in two or more patients (**Supplementary Table 7**).

Characterization of biological features. For LBEMs altering or not affecting splicing, we extracted the 9-nt 5' splice-site sequence (three bases in the exon and six bases in the intron) using BEDTools (getfasta module)⁵⁰. We estimated the splicing strength of the sequences with the wild-type and mutant alleles using two *in silico* methods: MaxEnt³³ and H-bond⁵⁹ scores. MaxEnt scores were calculated using a Perl script provided by the Burge laboratory. H-bond scores were calculated using the spliceSites R package, which required 11-nt 5' splice-site sequences as input (three bases in the exon and eight bases in the intron). The splicing strength of the 3' splice-site sequence was estimated using the MaxEnt method. Motif logos were generated using WebLogo⁶⁰. To investigate the density of splicing regulatory elements around SNVs, we first obtained 285 exonic⁶¹ and 400 intronic⁶² splicing regulatory motifs. To align the regulatory motifs against target regions, we used Burrows-Wheeler Aligner (BWA)⁶³ with default parameters but allowing exact matches and all multiple alignment. For each exon and intron, the fraction of bases covered by the exonic and intronic regulatory motifs was calculated.

Predictions based on classification models. We used the Lasso logistic regression model (R package glmnet)³⁸ to build three binary classifiers to distinguish somatic SNVs causing different types of abnormal splicing: (i) intron retention versus normal splicing, (ii) exon skipping versus normal splicing and (iii) intron retention versus exon skipping. The list of 18 features used for the prediction models is shown in **Supplementary Table 8**. We constructed the prediction models using a set of SNVs altering splicing identified by allele-specific splicing analysis. To evaluate unbiased classification performance and determine an optimal parameter (λ value), we iteratively excluded a single SNV and performed LOOCV on the remaining SNVs using the cv.glmnet module. We then tested the excluded SNV using a cross-validated model with the chosen λ value. The AUC was based on the pool of tested SNV results, and the AUC plot was generated using the R package pROC⁶⁴. The final λ value was determined by taking the average of the λ values from each LOOCV. With this value, the final prediction models were built using all of the input SNVs. We applied the models to LBEMs that had been excluded in the allele-specific splicing analysis for multiple reasons, such as a lack of mutant allele expression due to NMD.

To detect additional LBEMs associated with intron retention or exon skipping, we applied the prediction models to 474 LBEMs, including those whose

mutant allele expression was not observed, mainly because of NMD activation. To predict LBEMs associated with intron retention, both the intron retention versus normal and the intron retention versus exon skipping classifier must classify the mutations as causing intron retention. Likewise, to predict LBEMs associated with exon skipping, both the exon skipping versus normal and the intron retention versus exon skipping classifier must classify the mutations as causing exon skipping. LBEMs having a missing value in any of the genomic or transcriptomic features were excluded from prediction (**Supplementary Table 10**). For LBEMs in the first exon of tumor suppressors (for example, *CDKN2A*; **Table 1**), manual inspection of RNA-seq reads was performed because the computational model requires the expression change relative to the two adjacent exons and could not be applied.

Experimental validation using minigene splicing assays. For each region of interest, exonic and 90- to 130-bp flanking intronic sequences were amplified by PCR using the primers listed in **Supplementary Table 14**. To create mutant sequences, PCR-based site-directed mutagenesis was performed. Briefly, genomic DNA (US Biological) was first amplified by PCR using primers to generate two fragments with a 20- to 25-bp overlap containing a mutant site. Another round of PCR using nested primers then fused the fragments into a single molecule with the mutation. The wild-type and mutant sequences were cloned into the *ApaI* and *BglII* sites of pDUP4-1 (1) using DH5- α competent cells (Life Technologies). The inserted sequences replaced the 33-nt hybrid exon and its flanking intronic regions. Each construct verified by Sanger sequencing (3 μ g) was transiently transfected into HEK293FT cells using Lipofectamine 3000 (Life Technologies) in six-well tissue culture plates. Forty-eight hours after transfection, total RNA was isolated from the cells using the RNeasy mini kit (Qiagen). Total RNA (2 μ g) was treated with RNase-free DNase I (New England BioLabs) and used to synthesize cDNA with the SuperScript III First-Strand Synthesis System (Life Technologies) and 20-mer oligo(dT) primer. cDNA was amplified by PCR using the primers listed in **Supplementary Table 14** with Herculase II Fusion DNA polymerase (Agilent Technologies). The forward and reverse PCR primers corresponded to sequences of the target exon and *HBB* exon 2 of pDUP4-1, respectively. PCR products were separated by electrophoresis through a 2% agarose gel in 1 $\times$ TBE and stained with GelRed (Biotium). The absence of amplification in reverse transcriptase-negative controls was verified in all test sets. The ratio of splicing and intron retention isoforms was determined using ImageJ (US National Institutes of Health) and averaged from triplicates. To confirm the sequence of each band, the PCR products were gel purified using the QIAquick Gel Extraction kit (Qiagen) and verified by Sanger sequencing (**Supplementary Data Set**). HEK293FT cells were obtained from Invitrogen and regularly tested negative for mycoplasma contamination.

42. Cancer Genome Atlas Network. Comprehensive molecular portraits of human breast tumours. *Nature* **490**, 61–70 (2012).
43. Cancer Genome Atlas Research Network. Comprehensive molecular characterization of clear cell renal cell carcinoma. *Nature* **499**, 43–49 (2013).
44. Cancer Genome Atlas Research Network. Comprehensive genomic characterization of squamous cell lung cancers. *Nature* **489**, 519–525 (2012).
45. Cancer Genome Atlas Research Network. Integrated genomic analyses of ovarian carcinoma. *Nature* **474**, 609–615 (2011).
46. Kandoth, C. *et al.* Integrated genomic characterization of endometrial carcinoma. *Nature* **497**, 67–73 (2013).
47. Cancer Genome Atlas Network. Comprehensive molecular characterization of human colon and rectal cancer. *Nature* **487**, 330–337 (2012).
48. Karolchik, D. *et al.* The UCSC Genome Browser database: 2014 update. *Nucleic Acids Res.* **42**, D764–D770 (2014).
49. Pruitt, K.D. *et al.* RefSeq: an update on mammalian reference sequences. *Nucleic Acids Res.* **42**, D756–D763 (2014).
50. Quinlan, A.R. & Hall, I.M. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841–842 (2010).
51. Wu, T.D. & Watanabe, C.K. GMAP: a genomic mapping and alignment program for mRNA and EST sequences. *Bioinformatics* **21**, 1859–1875 (2005).
52. Benjamini, Y. & Hochberg, Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc., B* **57**, 289–300 (1995).
53. Yoshida, K. & Ogawa, S. Splicing factor mutations and cancer. *Wiley Interdiscip. Rev. RNA* (2014).
54. Lawrence, M.S. *et al.* Discovery and saturation analysis of cancer genes across 21 tumour types. *Nature* **505**, 495–501 (2014).

55. Futreal, P.A. *et al.* A census of human cancer genes. *Nat. Rev. Cancer* **4**, 177–183 (2004).
56. Solimini, N.L. *et al.* Recurrent hemizygous deletions in cancers may optimize proliferative potential. *Science* **337**, 104–109 (2012).
57. Zhao, M., Sun, J. & Zhao, Z. TSGene: a web resource for tumor suppressor genes. *Nucleic Acids Res.* **41**, D970–D976 (2013).
58. Kandoth, C. *et al.* Mutational landscape and significance across 12 major cancer types. *Nature* **502**, 333–339 (2013).
59. Freund, M. *et al.* A novel approach to describe a U1 snRNA binding site. *Nucleic Acids Res.* **31**, 6963–6975 (2003).
60. Crooks, G.E., Hon, G., Chandonia, J.M. & Brenner, S.E. WebLogo: a sequence logo generator. *Genome Res.* **14**, 1188–1190 (2004).
61. Fairbrother, W.G., Yeh, R.F., Sharp, P.A. & Burge, C.B. Predictive identification of exonic splicing enhancers in human genes. *Science* **297**, 1007–1013 (2002).
62. Wang, Z. *et al.* Systematic identification and analysis of exonic splicing silencers. *Cell* **119**, 831–845 (2004).
63. Li, H. & Durbin, R. Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics* **25**, 1754–1760 (2009).
64. Robin, X. *et al.* pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics* **12**, 77 (2011).